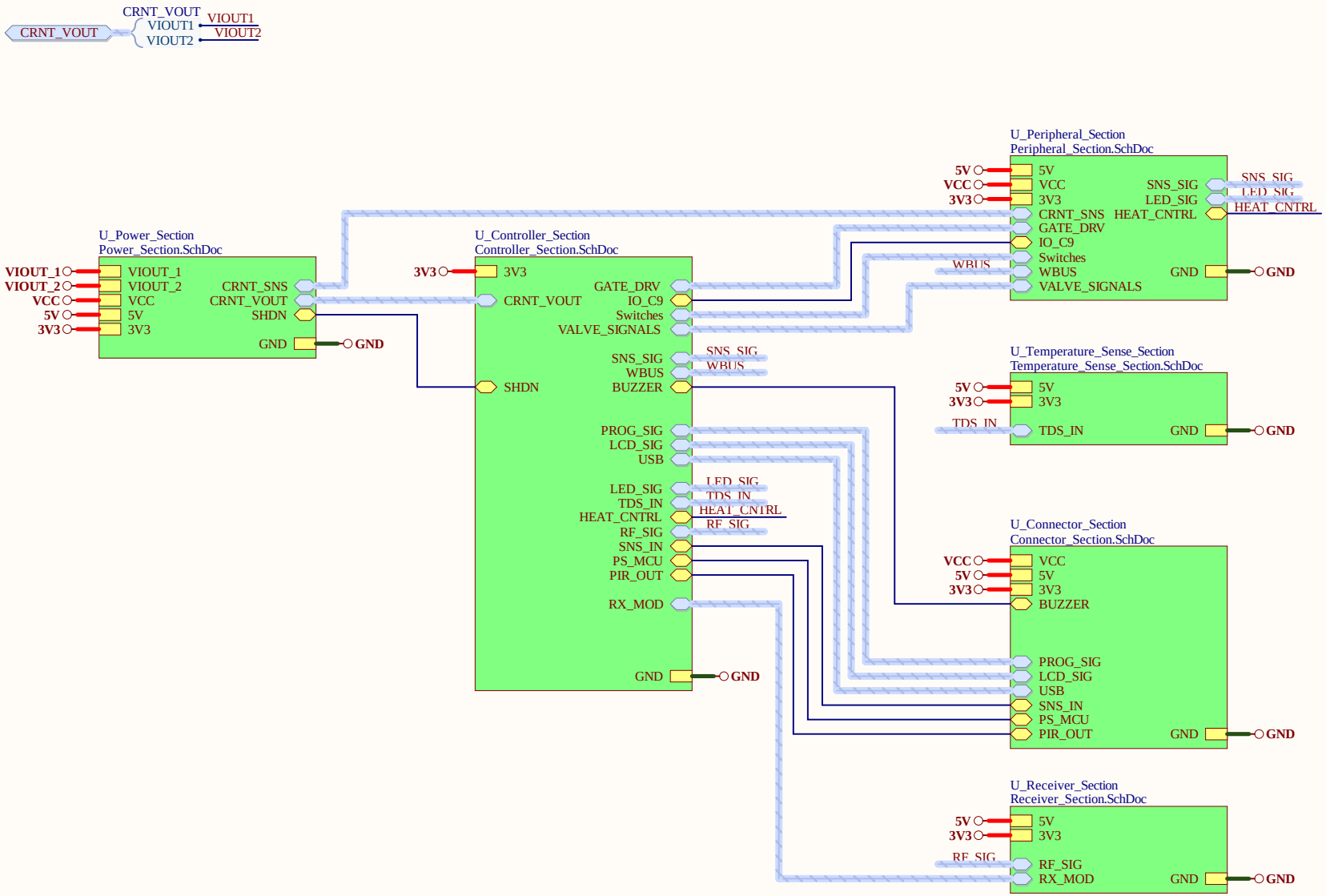
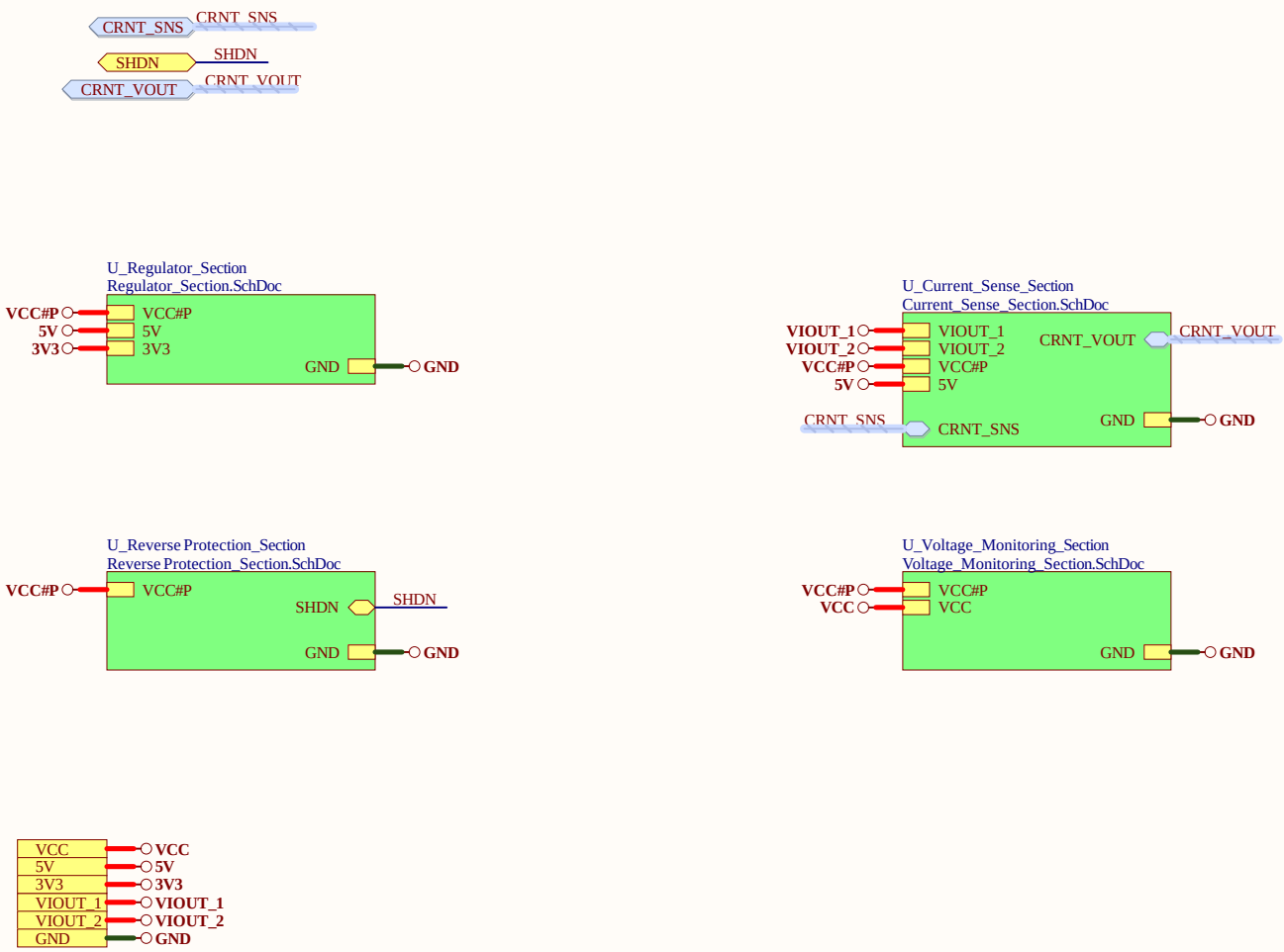
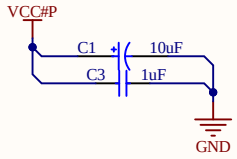

CRNT_VOUT
CRNT_VOUT
VIOUT1
VIOUT2
VIOUT1
VIOUT2
U_Power_Section
Power_Section.SchDoc
VIOUT_1
VIOUT_2
VCC
5V
3V3
VIOUT_1
VIOUT_2
VCC
5V
3V3
CRNT_SNS
CRNT_VOUT
SHDN
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
CRNT_VOUT
SHDN
GATE_DRV
IO_C9
Switches
VALVE_SIGNALS
SNS_SIG
WBUS
BUZZER
PROG_SIG
LCD_SIG
USB
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
SNS_IN
PS_MCU
PIR_OUT
RX_MOD
GND
GND
SNS_SIG
WBUS
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
U_Peripheral_Section
Peripheral_Section.SchDoc
5V
VCC
3V3
CRNT_SNS
GATE_DRV
IO_C9
Switches
WBUS
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
GND
GND
U_Temperature_Sense_Section
Temperature_Sense_Section.SchDoc
5V
3V3
TDS_IN
TDS_IN
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VCC
5V
3V3
BUZZER
PROG_SIG
LCD_SIG
USB
SNS_IN
PS_MCU
PIR_OUT
GND
GND
U_Receiver_Section
Receiver_Section.SchDoc
5V
3V3
RF_SIG
RF_SIG
RX_MOD
GND
GND

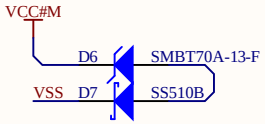

CRNT_SNS
CRNT_SNS
SHDN
SHDN
CRNT_VOUT
CRNT_VOUT
U_Regulator_Section
Regulator_Section.SchDoc
VCC#P
5V
3V3
VCC#P
5V
3V3
GND
GND
U_Current_Sense_Section
Current_Sense_Section.SchDoc
VIOUT_1
VIOUT_2
VCC#P
5V
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_VOUT
CRNT_VOUT
CRNT_SNS
CRNT_SNS
GND
GND
U_Reverse Protection_Section
Reverse Protection_Section.SchDoc
VCC#P
VCC#P
SHDN
SHDN
GND
GND
U_Voltage_Monitoring_Section
Voltage_Monitoring_Section.SchDoc
VCC#P
VCC
VCC#P
VCC
GND
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND



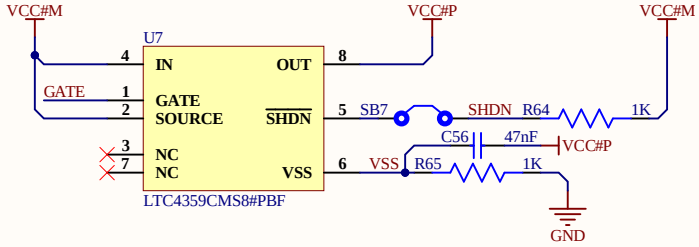
DECOUPLING CAPCITOR



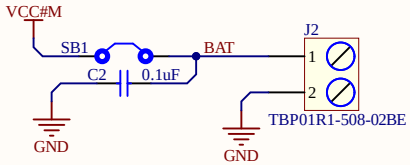
INPUT DIODES



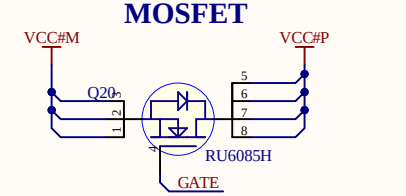
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT

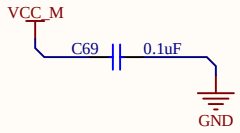


MOSFET

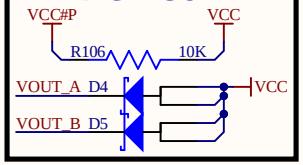


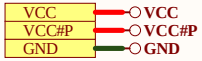


DECOUPLING CAPCITOR

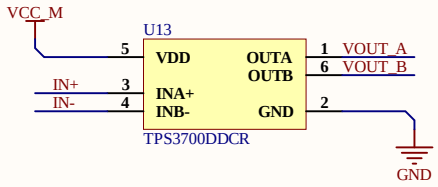


AND CIRCUITRY

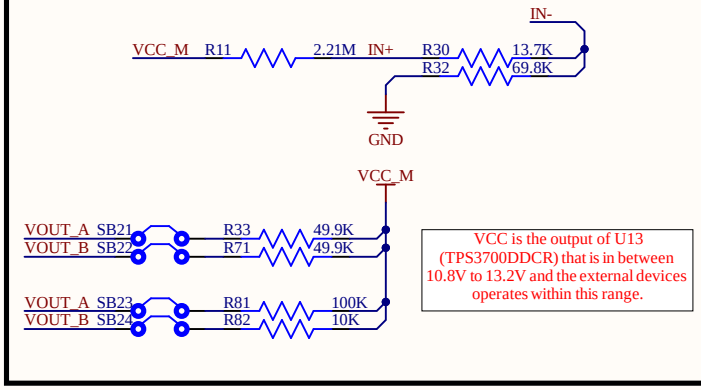




UNDER AND OVER VOLTAGE PROTECTION IC



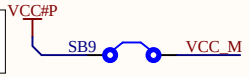
SUPPORTING CIRCUITRY



VCC is the output of U13 (TPS3700DDCR) that is in between 10.8V to 13.2V and the external devices operates within this range.

VOLTAGE TO MONITOR

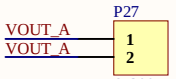
Monitors Main 12V Input. Need 0R Resistor or Shorting Jumper to provide connections to Solder Bridge (SB9).



UNDER VOLTAGE OUTPUT

Do Not Mount

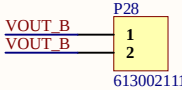
Under Voltage Open Drain Output. 100K Pullup is recommended.

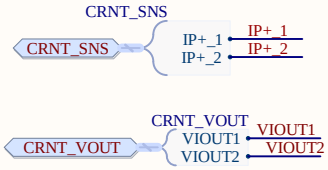


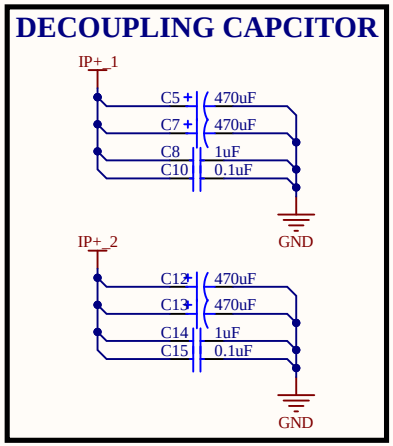
OVER VOLTAGE OUTPUT

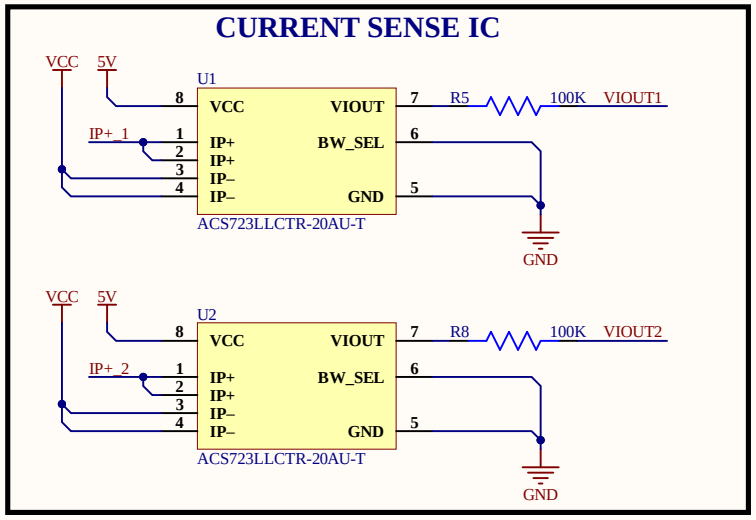
Do Not Mount

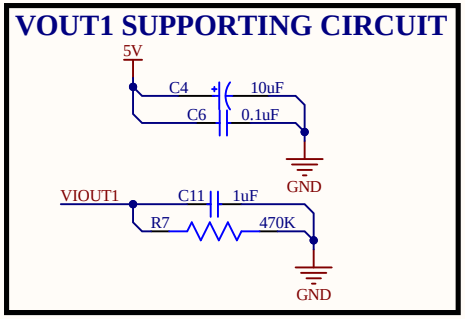
Over Voltage Open Drain Output. 10K Pullup is recommended.

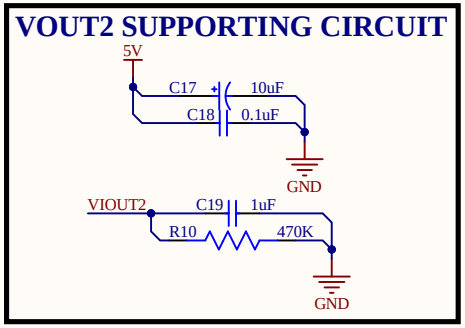




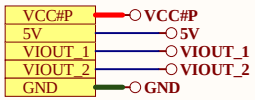




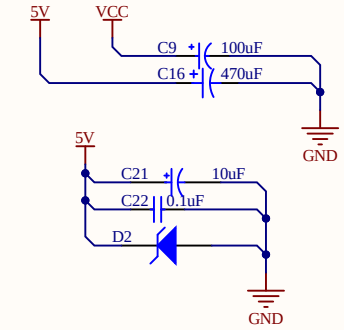


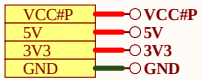


Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

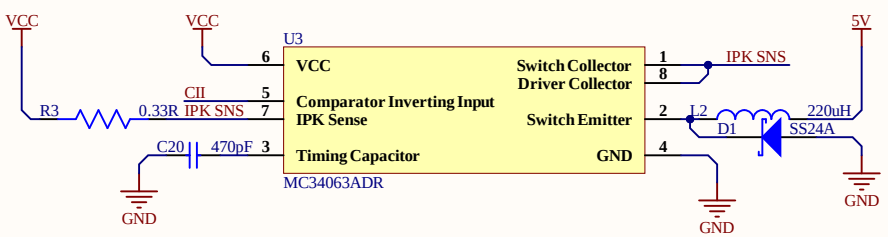


DECOUPLING CAPCITOR

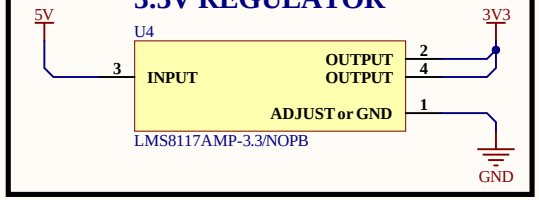




5V BUCK REGULATOR

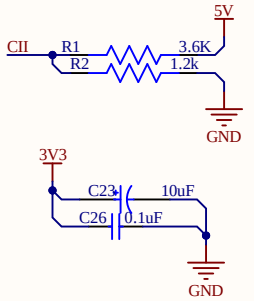


3.3V REGULATOR

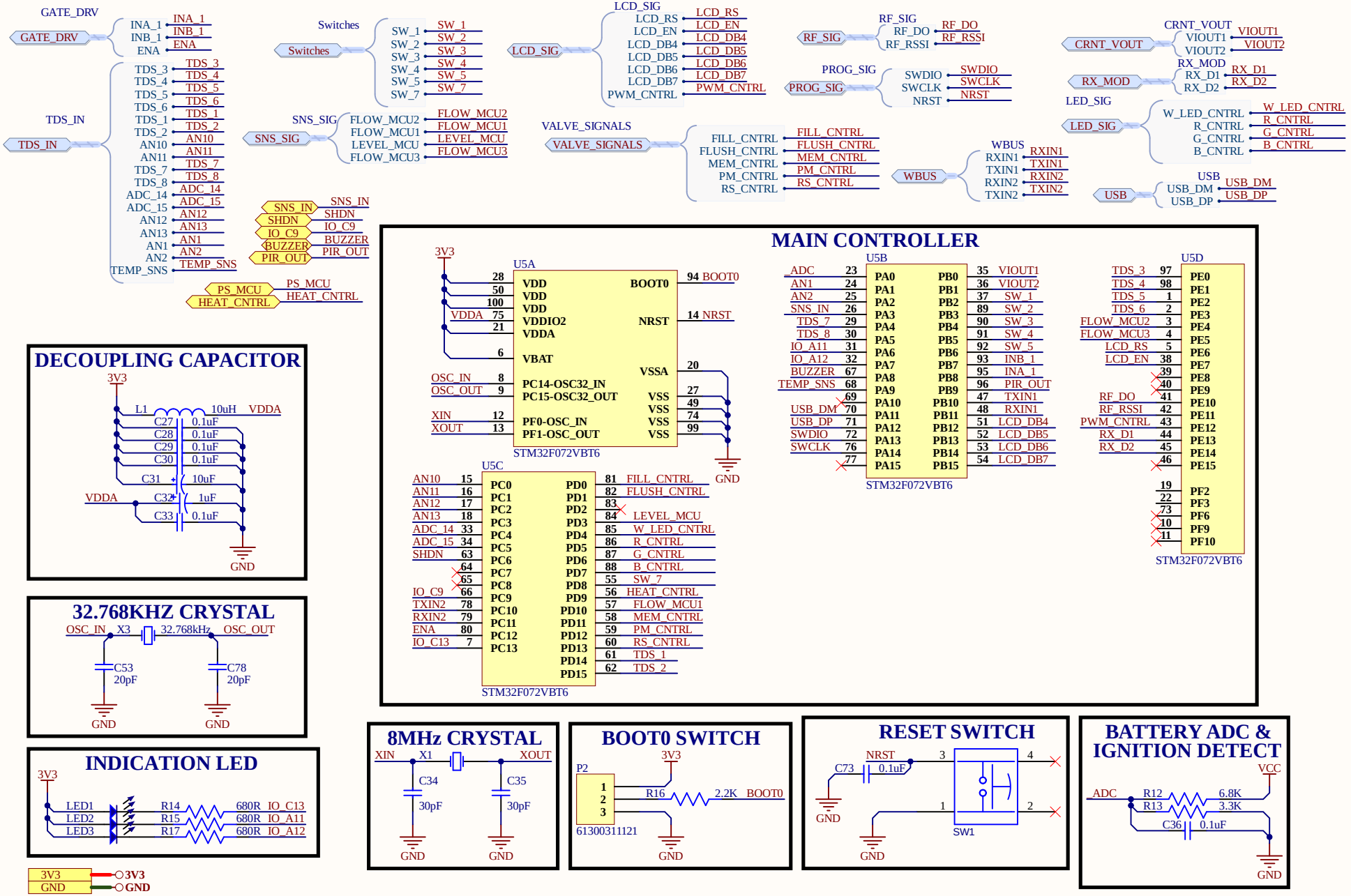


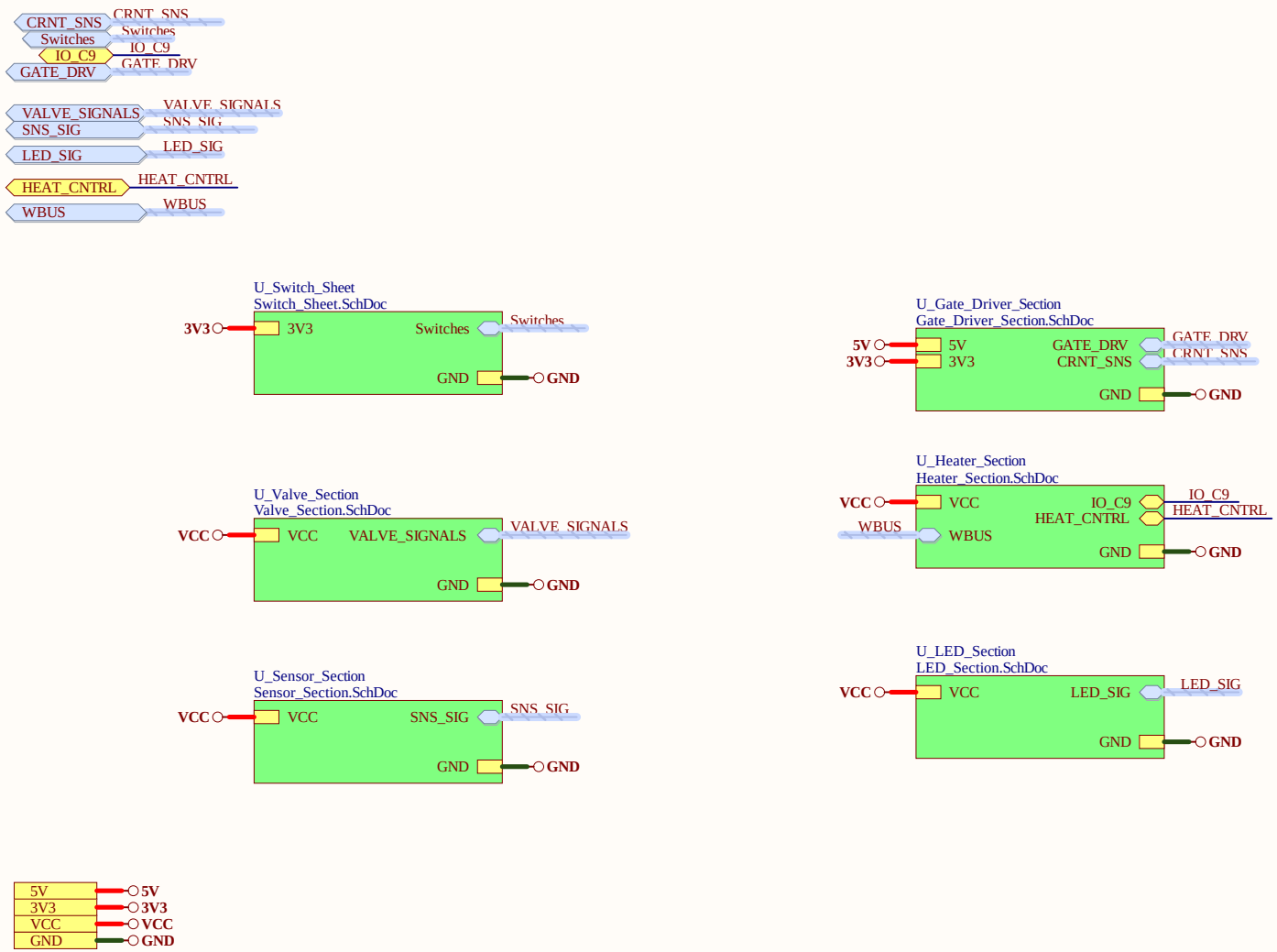
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

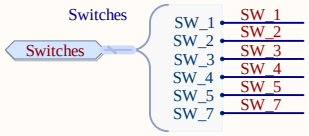
SUPPORTING CIRCUIT

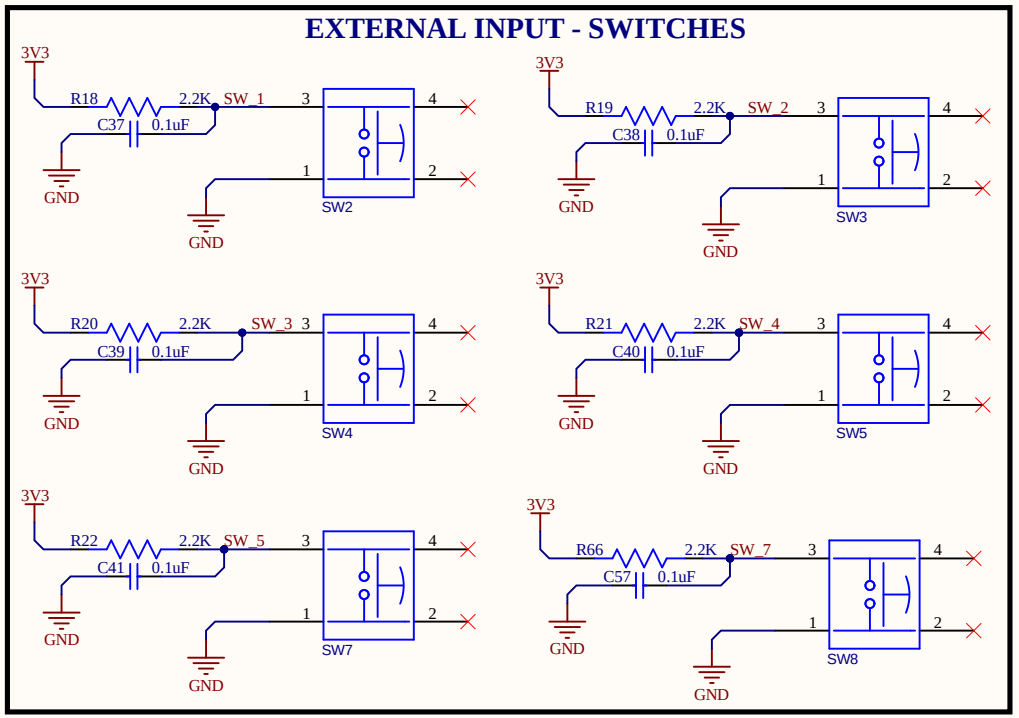


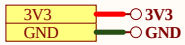


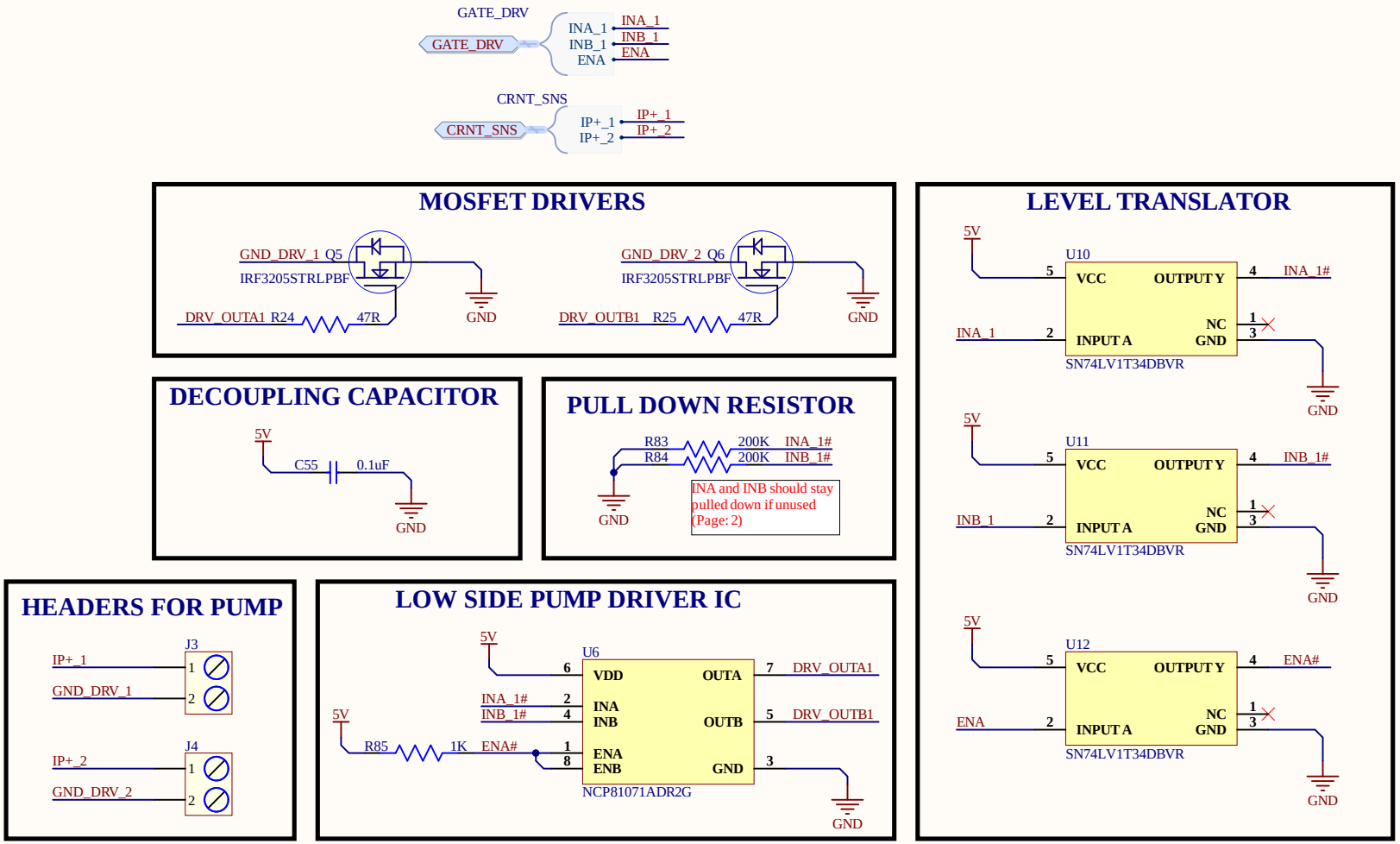

GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
TEMP_SNS
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SNS_SIG
FLOW_MCU2
FLOW_MCU1
LEVEL_MCU
FLOW_MCU3
LCD_SIG
LCD_RS
LCD_EN
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
PWM_CNTRL
RF_SIG
RF_DO
RF_RSSI
PROG_SIG
SWDIO
SWCLK
NRST
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
CRNT_VOUT
VIOUT1
VIOUT2
RX_MOD
RX_D1
RX_D2
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
USB
USB_DM
USB_DP
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
MAIN CONTROLLER
U5A
STM32F072VBT6
U5B
U5C
U5D
DECOUPLING CAPACITOR
3V3
L1
10uH
VDDA
C27
C28
C29
C30
C31
C32
C33
0.1uF
10uF
1uF
GND
32.768KHZ CRYSTAL
OSC_IN
X3
32.768kHz
OSC_OUT
C53
C78
20pF
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R
8MHz CRYSTAL
XIN
X1
XOUT
C34
C35
30pF
BOOT0 SWITCH
P2
R16
2.2K
BOOT0
61300311121
RESET SWITCH
NRST
C73
SW1
BATTERY ADC & IGNITION DETECT
VCC
ADC
R12
R13
6.8K
3.3K
C36

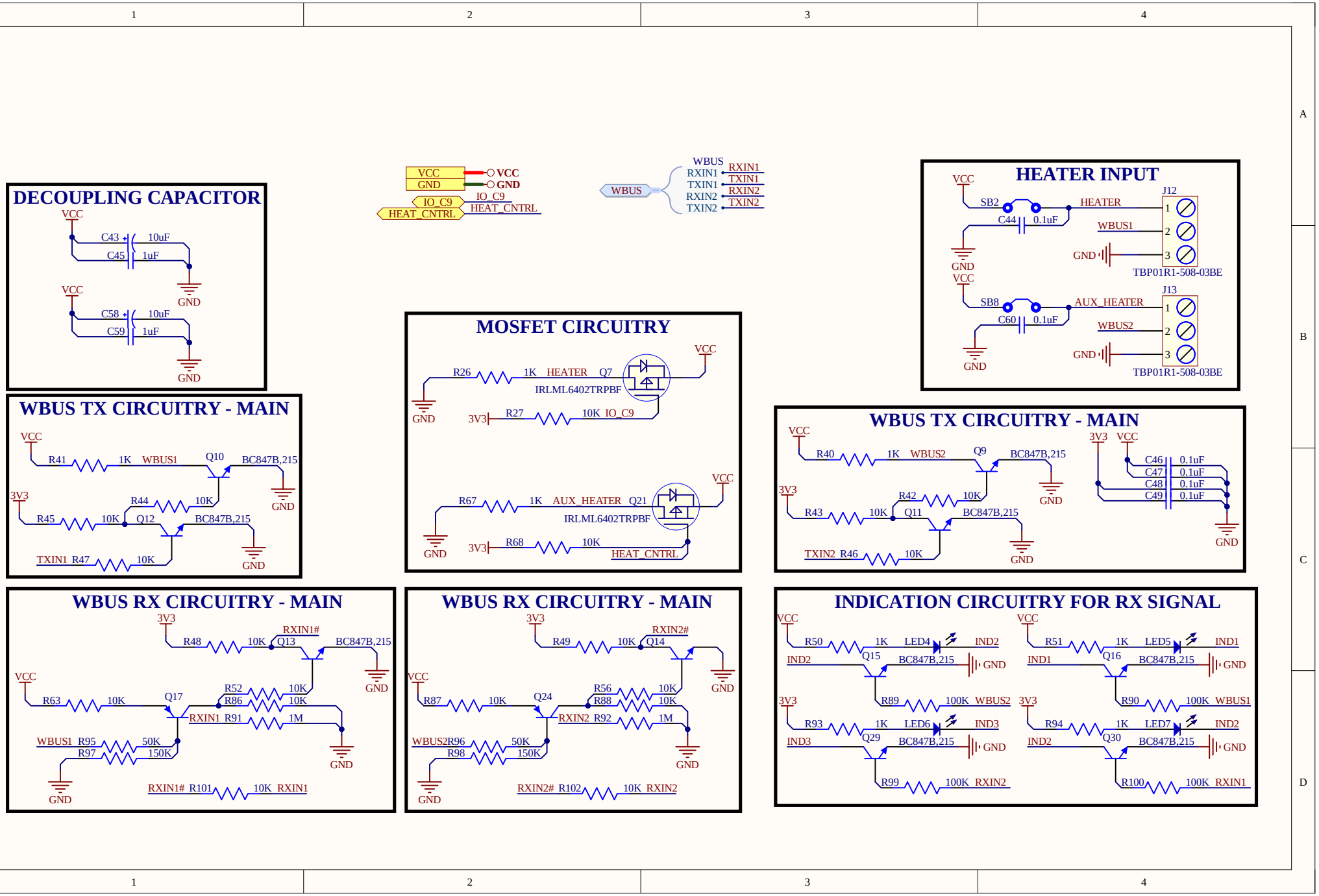

CRNT_SNS
Switches
IO_C9
GATE_DRV
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
WBUS
U_Switch_Sheet
Switch_Sheet.SchDoc
3V3
Switches
GND
U_Gate_Driver_Section
Gate_Driver_Section.SchDoc
5V
3V3
GATE_DRV
CRNT_SNS
GND
U_Valve_Section
Valve_Section.SchDoc
VCC
VALVE_SIGNALS
U_Heater_Section
Heater_Section.SchDoc
IO_C9
HEAT_CNTRL
WBUS
U_Sensor_Section
Sensor_Section.SchDoc
SNS_SIG
U_LED_Section
LED_Section.SchDoc
LED_SIG


Switches
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7


EXTERNAL INPUT - SWITCHES
3V3
R18
2.2K
SW_1
C37
0.1uF
GND
SW2
R19
SW_2
C38
SW3
R20
SW_3
C39
SW4
R21
SW_4
C40
SW5
R22
SW_5
C41
SW7
R66
SW_7
C57
SW8


3V3
GND
3V3
GND


GATE_DRV
GATE_DRV
INA_1
INB_1
ENA
INA_1
INB_1
ENA
CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2
MOSFET DRIVERS
GND_DRV_1 Q5
IRF3205STRLPBF
DRV_OUTA1 R24
47R
GND
GND_DRV_2 Q6
IRF3205STRLPBF
DRV_OUTB1 R25
47R
GND
LEVEL TRANSLATOR
5V
U10
5
VCC
OUTPUT Y
4
INA_1#
INA_1
2
INPUT A
NC
GND
1
3
SN74LV1T34DBVR
GND
5V
U11
5
VCC
OUTPUT Y
4
INB_1#
INB_1
2
INPUT A
NC
GND
1
3
SN74LV1T34DBVR
GND
5V
U12
5
VCC
OUTPUT Y
4
ENA#
ENA
2
INPUT A
NC
GND
1
3
SN74LV1T34DBVR
GND
DECOUPLING CAPACITOR
5V
C55
0.1uF
GND
PULL DOWN RESISTOR
R83
200K
INA_1#
R84
200K
INB_1#
GND
INA and INB should stay pulled down if unused (Page: 2)
HEADERS FOR PUMP
J3
IP+_1
1
GND_DRV_1
2
J4
IP+_2
1
GND_DRV_2
2
LOW SIDE PUMP DRIVER IC
5V
U6
6
VDD
OUTA
7
DRV_OUTA1
INA_1#
2
INA
INB_1#
4
INB
OUTB
5
DRV_OUTB1
5V
R85
1K
ENA#
1
8
ENA
ENB
GND
3
NCP81071ADR2G
GND
3V3
5V
GND
3V3
5V
GND


DECOUPLING CAPACITOR
VCC
C43 10uF
C45 1uF
GND
VCC
C58 10uF
C59 1uF
GND
VCC
GND
IO_C9
HEAT_CNTRL
VCC
GND
IO_C9
HEAT_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
HEATER INPUT
VCC
SB2
C44 0.1uF
GND
HEATER
WBUS1
GND
J12
1
2
3
TBP01R1-508-03BE
VCC
SB8
C60 0.1uF
GND
AUX_HEATER
WBUS2
GND
J13
1
2
3
TBP01R1-508-03BE
MOSFET CIRCUITRY
VCC
R26 1K HEATER Q7
IRLML6402TRPBF
GND
3V3 R27 10K IO_C9
VCC
R67 1K AUX_HEATER Q21
IRLML6402TRPBF
GND
3V3 R68 10K HEAT_CNTRL
WBUS TX CIRCUITRY - MAIN
VCC
R41 1K WBUS1 Q10 BC847B,215
GND
3V3
R44 10K
R45 10K Q12 BC847B,215
GND
TXIN1 R47 10K
WBUS TX CIRCUITRY - MAIN
VCC
R40 1K WBUS2 Q9 BC847B,215
GND
3V3
R42 10K
R43 10K Q11 BC847B,215
GND
TXIN2 R46 10K
3V3 VCC
C46 0.1uF
C47 0.1uF
C48 0.1uF
C49 0.1uF
GND
WBUS RX CIRCUITRY - MAIN
3V3
R48 10K Q13 BC847B,215
RXIN1#
GND
VCC
R63 10K Q17
R52 10K
R86 10K
RXIN1 R91 1M
GND
WBUS1 R95 50K
R97 150K
GND
RXIN1# R101 10K RXIN1
WBUS RX CIRCUITRY - MAIN
3V3
R49 10K Q14 BC847B,215
RXIN2#
GND
VCC
R87 10K Q24
R56 10K
R88 10K
RXIN2 R92 1M
GND
WBUS2R96 50K
R98 150K
GND
RXIN2# R102 10K RXIN2
INDICATION CIRCUITRY FOR RX SIGNAL
VCC
R50 1K LED4 IND2
IND2
Q15 BC847B,215 GND
3V3
R89 100K WBUS2
R93 1K LED6 IND3
IND3
Q29 BC847B,215 GND
R99 100K RXIN2
VCC
R51 1K LED5 IND1
IND1
Q16 BC847B,215 GND
3V3
R90 100K WBUS1
R94 1K LED7 IND2
IND2
Q30 BC847B,215 GND
R100 100K RXIN1

VALVE_SIGNALS

VALVE_SIGNALS

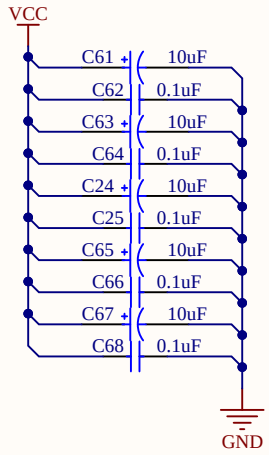
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

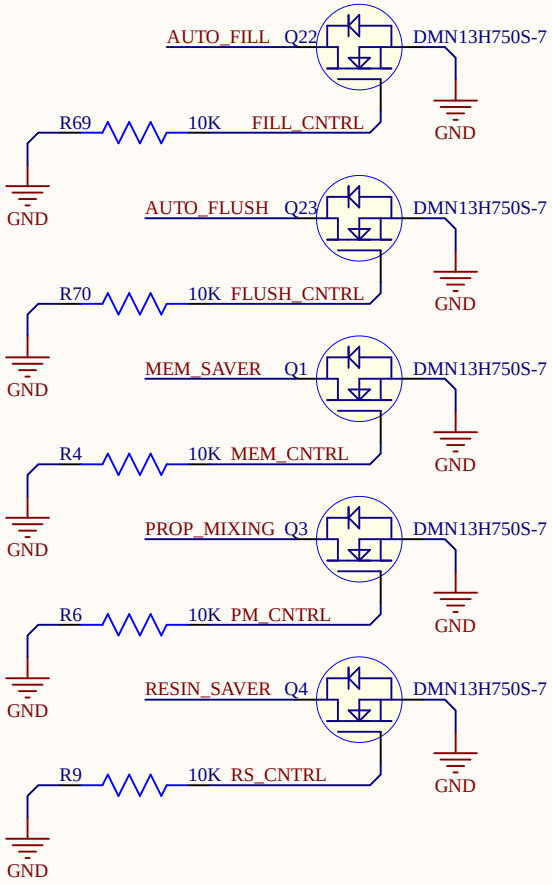
VCC
GND

VCC
GND

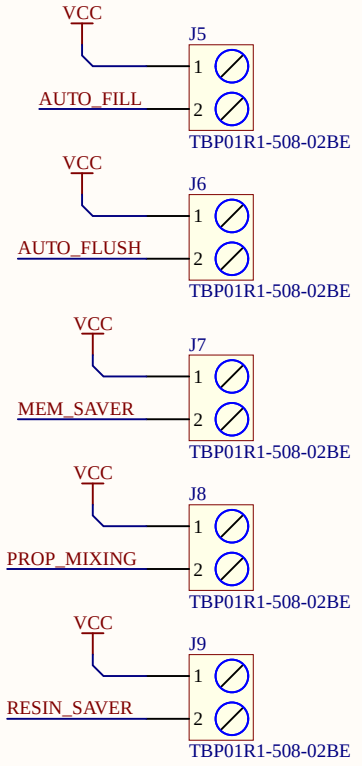
DECOUPLING CAPACITOR



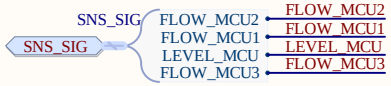
SOLENOID CONTROL CIRCUIT



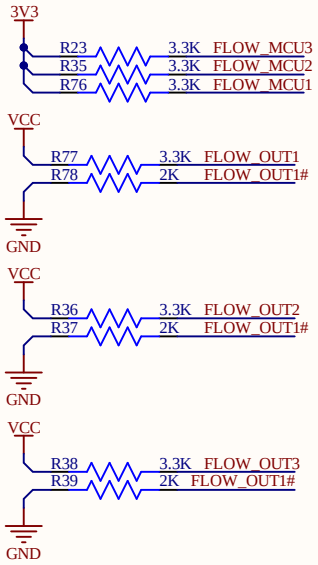
SOLENOID VALVE INPUTS



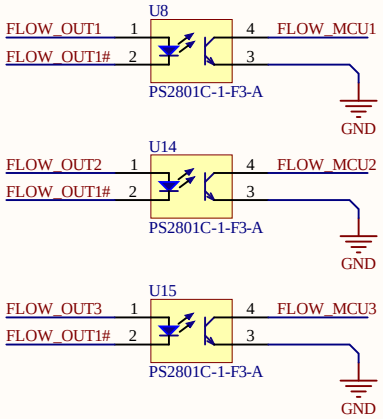
Title		
Size A	Number	Revision
Date: 8/07/2022		Sheet of
File: C:\Users\..\Valve_Section.SchDoc		Drawn By:



PULL UP RESISTORS

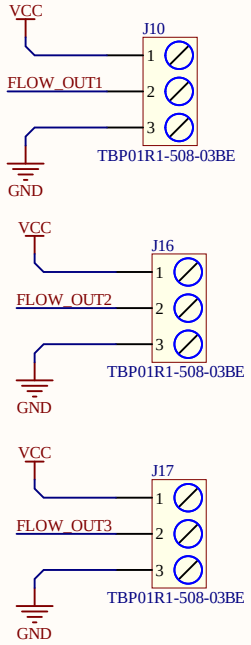


OPTO ISOLATION FOR FLOW METER

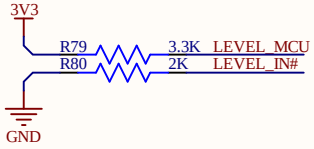


Forward voltage of diode=1.2V
With VCC=12V, IF=5.4mA
Flow meter datasheet recommends a pull up value between 3K-30K

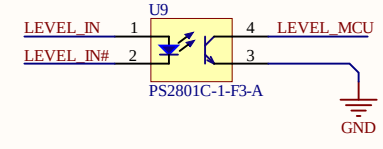
FLOW METER INPUT



SUPPORTING CIRCUIT

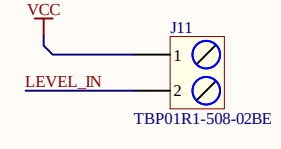


OPTO ISOLATION FOR FLOW METER



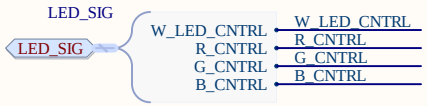
Forward voltage of diode=1.2V
With VCC=12V, IF=54mA

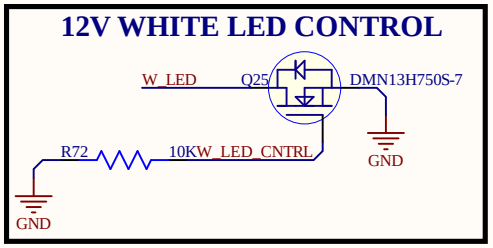
LEVEL SENSOR INPUT

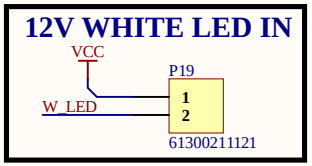


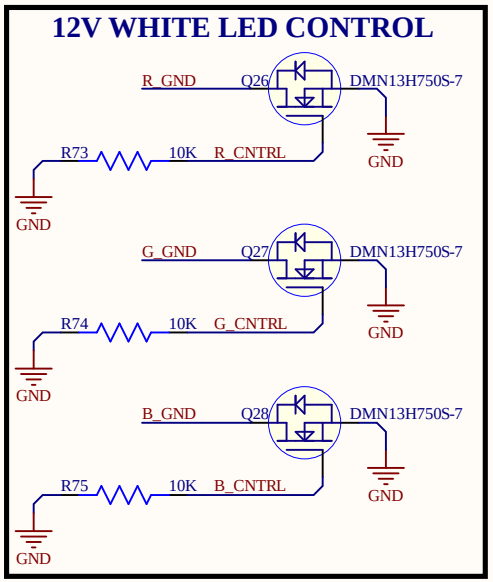


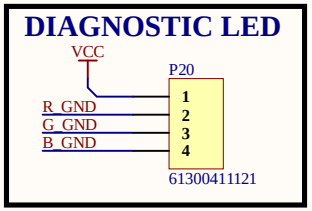
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Size: A4	Number	Revision
Date: 8/07/2022		Sheet of
File: C:\Users\..\Sensor_Section.SchDoc		Drawn By:





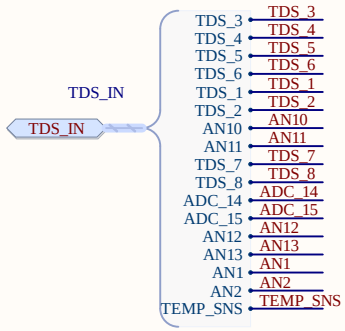




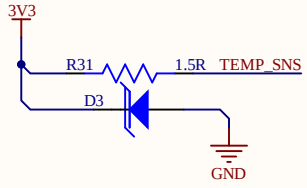




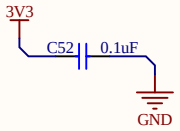
Title		
Size: A4	Number	Revision
Date: 8/07/2022		Sheet of
File: C:\Users\..\LED_Section.SchDoc		Drawn By:

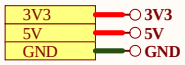


PULL UP RESISTORS

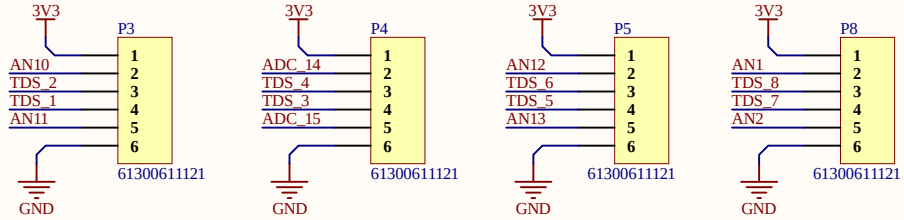


DECOUPLING CAP



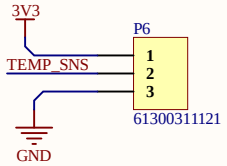


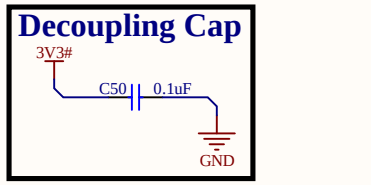
4 X TDS INPUTS

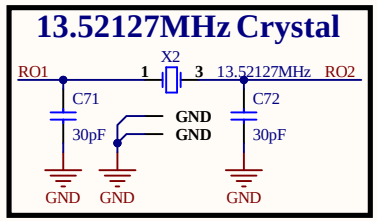


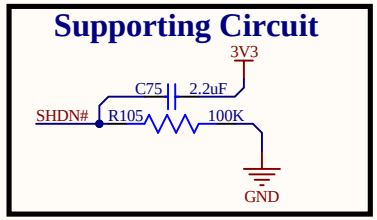
These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

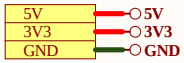
TEMP SENSOR DS18B20





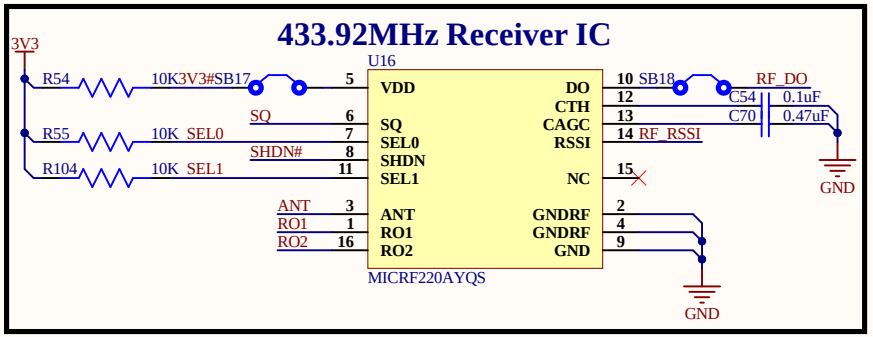




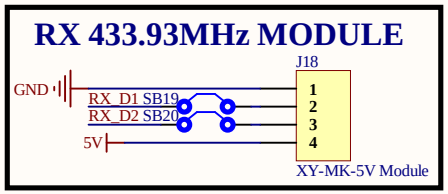


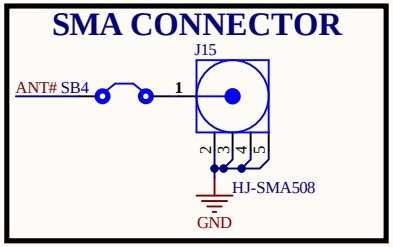


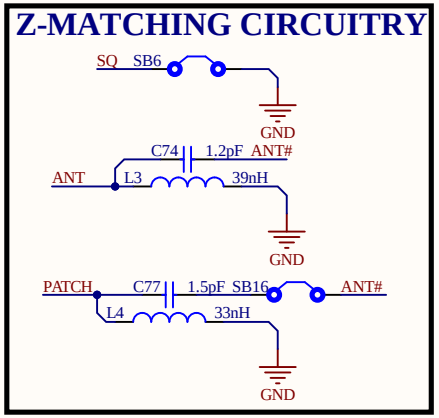




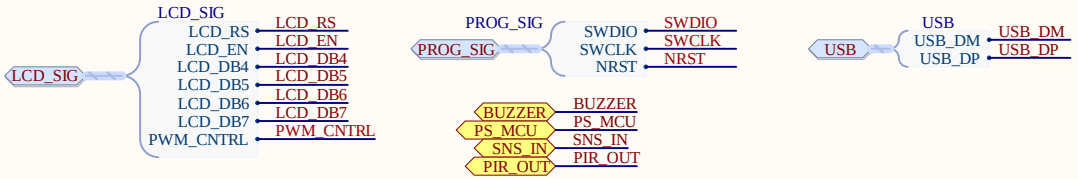
When SEL0 & SEL1 are pulled up, low pass filter BW = 13000Hz
Max encoded bit rate =20Kbps
Pulse width @433.92MHz=38us
CAGC Valu= 0.47uF
FREF= FRF/(32+87/1000)
FREF= 13.52323MHz (@433.92MHz FRF)



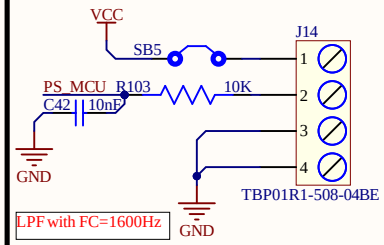




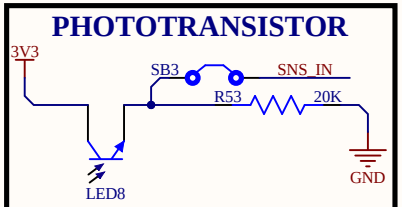
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: V3_Board.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/07/2022	Sheet 16 of 17	
	File: C:\Users\..\Receiver_Section.SchDoc	Drawn By: Huzaifa Ali	



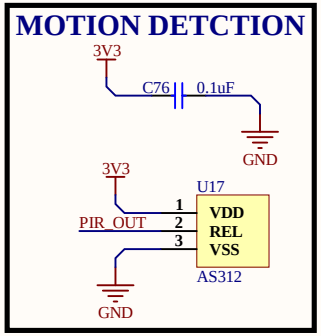
PRESSURE SENSOR



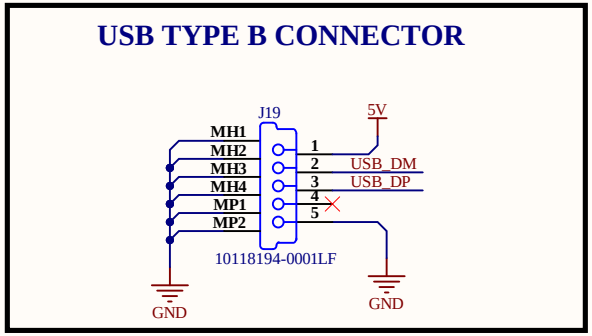
PHOTOTRANSISTOR



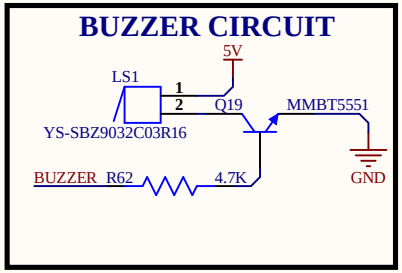
MOTION DETCTION



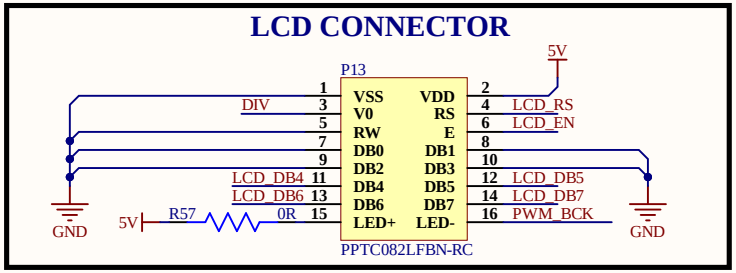
USB TYPE B CONNECTOR



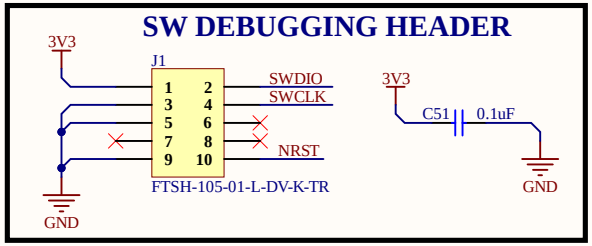
BUZZER CIRCUIT



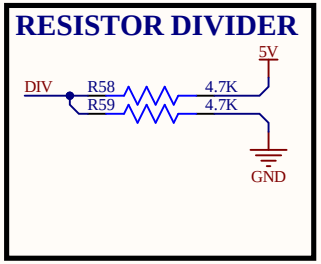
LCD CONNECTOR



SW DEBUGGING HEADER



RESISTOR DIVIDER



PWM BACKLIGHT

